

Notes: Macchiavello and Morjaria (QJE, 2021)

Competition and Relational Contracts in the Rwanda Coffee Chain

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1 Big picture

- **Question.** How does increased competition affect market outcomes when formal contracts are hard to enforce and firms rely on relational contracts?
- **Context.** The paper studies Rwanda’s coffee sector, where downstream washing stations or mills buy coffee cherries from upstream smallholder farmers.
- **Relational contract.** Mills and farmers often exchange legally unenforceable services across the season: inputs or loans before harvest, cherries sold on credit during harvest, and postharvest assistance unrelated to immediate operations.
- **Identification challenge.** Competition is endogenous because mills may enter where market conditions, farmer quality, roads, tree density, or local capacity already differ.
- **Instrument.** The authors build an instrument from an engineering model for the optimal placement of coffee washing stations. Conditional on suitability inside the mill’s catchment area, suitability in the surrounding donut area predicts competing mills but should not directly affect the focal mill except through competition.
- **Main result.** An additional competing mill reduces the relational-contract index by about 0.28 standard deviations in the IV specification.
- **Performance result.** Competition worsens mill outcomes: it reduces relational practices, raises operating frictions, lowers quality, and reduces the amount of coffee processed by mills.
- **Farmer result.** Competition makes farmers worse off in this setting. It reduces aggregate cherries supplied to mills and weakens interlinked services that farmers value.
- **Mechanism.** Competition breaks relational contracts through two channels: directly, by raising farmers’ outside options and temptation to default, and indirectly, by reducing mills’ profits and hence their future relationship surplus.
- **Takeaway.** In weak institutional environments, more competition is not always welfare improving. Competition can destroy relational governance and reduce market efficiency.

2 Incremental contribution of this paper

2.1 Contribution relative to the competition-productivity literature

- A large empirical literature shows that competition often improves firm productivity and management by disciplining firms and selecting out inefficient producers.
- This paper shows an important boundary condition: when trade depends on relational contracts, competition can reduce efficiency by undermining the future rents that sustain cooperation.
- The contribution is not simply that competition may have negative effects. It identifies a concrete organizational mechanism: the weakening of relational contracting between upstream producers and downstream processors.

Interpretation: The paper moves the competition debate from a first-best market logic to a second-best institutional logic. In markets where formal enforcement is weak, rents may be productive because they sustain cooperation.

2.2 Contribution relative to relational-contract theory

- Relational-contract theory predicts that outside options and continuation values determine whether informal agreements can be sustained.

- Empirical tests are difficult because relational contracts are implicit, context-specific, and rarely observed directly.
- This paper constructs direct survey measures of relational practices and links them to exogenous variation in competition.

Interpretation: The paper is unusual because it observes both sides of the relationship: the mill's reported practices and farmers' reported experience of those practices.

2.3 Contribution relative to agricultural supply-chain studies

- Development and agricultural studies often emphasize that competition among buyers can improve prices for farmers.
- The paper shows that when transactions are bundled with services, inputs, credit, and quality incentives, spot-market competition may reduce the value of the trading relationship.
- The paper therefore contributes to understanding why agricultural supply chains often rely on stable buyer-seller relationships rather than purely anonymous spot markets.

Interpretation: The key insight is that the relevant object is not just the farmgate price. The total contract includes working capital, inputs, credit, quality support, and insurance-like assistance.

2.4 Contribution relative to empirical identification of market structure

- The paper develops a spatial instrument based on an engineering entry model for mills.
- Suitability inside the catchment area is controlled for, while suitability in the surrounding donut area instruments for competition.
- This provides a credible way to separate the focal mill's own location fundamentals from the presence of nearby competitors.

Interpretation: The identification design is a major contribution because it turns geographic suitability into a source of variation in market structure while conditioning on the focal market's own suitability.

3 Data and measurement

3.1 Rwanda coffee sector

- Rwanda's coffee sector consists of smallholder farmers producing cherries and mills processing cherries into parchment coffee.
- Mills must source fresh cherries during the harvest season. Cherries are perishable and must be processed quickly.
- Farmers often require inputs, finance, agronomic advice, and liquidity before harvest.
- Formal enforcement is weak, so many exchanges are sustained through repeated relationships.

Interpretation: The sector is ideal for studying relational contracting because trade is repeated, services are bundled with sourcing, and spot-market alternatives exist but are imperfect.

3.2 Mill survey

- The authors survey mill managers and obtain information on operational characteristics, capacity, production, costs, procurement, and relational practices.
- Table I reports that the average mill is about four years old and has theoretical capacity around 423 tons of cherries.
- The average mill faces about 6.5 competing mills within 10 km.
- Survey measures include inputs provided to farmers, help or loans to farmers, whether farmers sell on credit, and other relationship-based practices.

Interpretation: Mill survey data are the core source for measuring both relational contracting and operational outcomes.

3.3 Farmer survey

- The farmer survey is conducted among farmers around surveyed mills.
- It records farmer demographics, cooperative membership, distance to mill, coffee sales, prices, job satisfaction, and whether the farmer receives specific services from the mill.
- Table I reports that the average farmer is about 46 years old, has about five years of schooling, and lives about 5.5 km from the mill.

Interpretation: Farmer survey data provide independent validation of mill-reported relational practices and allow the authors to study farmer welfare and behavior.

3.4 Relational-contract index

The paper constructs relational-contract measures using indicators for:

- mill-provided inputs before harvest;
- credit and delayed payment during harvest;
- help, loans, and postharvest assistance unrelated to harvest operations.

The main relational-contract index is an equally weighted index of standardized farmer- and mill-manager-based indicators:

$$RC_m = \frac{1}{K} \sum_{k=1}^K z_{mk}, \quad (1)$$

where z_{mk} denotes the standardized value of relational-contract practice k for mill m .

Interpretation: The index aggregates multiple practices because the theory predicts complementarities. Competition should weaken the relationship as a system, not merely one isolated practice.

3.5 Competition measure

The baseline competition measure is the number of mills within 10 km of the focal mill:

$$C_m = \#\{\text{mills within 10 km of mill } m\}. \quad (2)$$

- The 10 km measure is conservative and corresponds to overlapping catchment areas.
- Mills report similar competition intensity in survey responses.
- Alternative measures are explored in the online appendix.

Interpretation: Competition is spatial because farmers can sell cherries to nearby mills, but perishable cherries limit the feasible radius.

4 Theoretical framework

4.1 Players and timing

- A mill repeatedly interacts with many farmers.
- Before harvest, the mill may provide inputs, loans, or other support.
- During harvest, the farmer chooses whether to deliver cherries to the mill or sell elsewhere.
- The mill then processes cherries and may continue or exit in future periods.

Interpretation: The contract is relational because the mill's preharvest support and the farmer's later delivery choices cannot be fully enforced by courts.

4.2 Relational-contract incentive constraints

A farmer honors the relational contract if the value of staying with the mill exceeds the gain from defaulting to an outside buyer:

$$\text{Continuation value from relationship} \geq \text{spot-market deviation gain.} \quad (3)$$

A mill provides services if the discounted future profits from the relationship exceed the current costs of those services:

$$\text{Future sourcing surplus} \geq \text{current cost of inputs, credit, and support.} \quad (4)$$

Interpretation: Competition can relax or tighten each side's constraint depending on prices and future rents. The paper emphasizes two forces that tighten the constraints.

4.3 Direct effect of competition

Competition raises the farmer's outside option. If there are more nearby mills, the farmer can more easily take inputs or credit from the focal mill and then sell cherries to a different mill.

$$C_i \uparrow \Rightarrow \text{farmer deviation gain} \uparrow. \quad (5)$$

Interpretation: This direct channel predicts that the negative effect of competition should be stronger for farmers closer to alternative mills or with better alternative access.

4.4 Indirect effect of competition

Competition reduces the focal mill's profits. Lower profits reduce the future surplus available to sustain relational contracts.

$$C_m \uparrow \Rightarrow \Pi_m \downarrow \Rightarrow \text{relationship surplus} \downarrow. \quad (6)$$

Interpretation: This indirect channel predicts that competition from mills a farmer cannot directly sell to may still weaken the relationship by reducing the focal mill's profits.

4.5 Testable predictions

1. Competition reduces the use of relational-contract practices.
2. Relational practices decline jointly because the practices are complementary.
3. Competition worsens mill operations by increasing irregular sourcing, costs, and quality problems.
4. Competition reduces farmer welfare if lost relational services dominate any spot-market benefits.
5. Farmer-level heterogeneity should distinguish direct and indirect mechanisms.

5 Empirical specifications

5.1 OLS specification

The baseline OLS regression is:

$$RC_m = \alpha + \beta C_m + X'_m \gamma + Z'_m \delta + \varepsilon_m, \quad (7)$$

where C_m is the number of mills within 10 km, X_m includes mill controls, and Z_m includes geographic controls.

Interpretation: The OLS coefficient is informative but potentially biased because competition is endogenous and measured with error.

5.2 Engineering model instrument

The instrument is based on suitability for mill placement in the donut area 5–10 km around the focal mill. The first stage is:

$$C_m = \alpha + \beta \widehat{S}_m^{5/10} + \rho S_m^{0/5} + X'_m \gamma + Z'_m \delta + u_m, \quad (8)$$

where:

- $\widehat{S}_m^{5/10}$ is average predicted suitability in the surrounding 5–10 km donut;
- $S_m^{0/5}$ controls for suitability inside the focal catchment area;
- X_m and Z_m include mill and geographic controls.

Interpretation: The key exclusion restriction is that, conditional on inside-catchment suitability, outside-donut suitability affects the focal mill only through nearby competitor entry.

5.3 Second stage

The IV specification is:

$$RC_m = \alpha + \beta \widehat{C}_m + X'_m \gamma + Z'_m \delta + \varepsilon_m. \quad (9)$$

Interpretation: The coefficient β estimates the causal effect of one additional competing mill on relational contracting.

5.4 Farmer-level mechanism specification

The farmer-level regressions use outcomes such as farmer welfare, relationship index, and sales decisions. A simplified specification is:

$$y_{im} = \alpha + \beta_1 C_m + \beta_2 \text{AlternativeAccess}_{im} + X'_{im} \gamma + \eta_m + \varepsilon_{im}. \quad (10)$$

Interpretation: Farmer-level tests separate the direct channel from the indirect mill-profit channel by exploiting distance and access to alternative mills.

6 Identification and empirical design

6.1 Why competition is endogenous

- Mills may enter places with better roads, more coffee trees, better terrain, or better farmer networks.
- These same fundamentals can affect relational contracting and mill performance.
- Good relational mills could attract competitors, or poor mills might create entry opportunities for competitors.
- Spatial competition is also measured with error if farmers' actual feasible trading areas differ from the assumed catchment radius.

Interpretation: These concerns mean simple cross-sectional correlations between competition and relational practices are not enough.

6.2 Why the instrument is plausible

- Engineering suitability predicts where mills are physically likely to be placed.
- Suitability in the focal catchment area is controlled directly.
- The excluded instrument is suitability outside the catchment, where it affects the focal mill through competitor entry.
- Robustness tests examine roads, tree density, alternative catchment sizes, and alternative entry-model specifications.

Interpretation: The design is a spatial shift-share style logic: outside areas generate competition pressure, while inside suitability controls for local fundamentals.

6.3 Interpretation of LATE

The IV estimates identify the effect of an additional competitor induced by exogenous geographic suitability for mill entry in the donut area. The average mill already faces several competitors.

Interpretation: The estimates should not be interpreted as the effect of moving from monopoly to competition. They identify the marginal effect of an additional competitor in a market with about six competitors on average.

6.4 Limitations

- The analysis is cross-sectional and based on one sector and one country.
- Relational contracts are measured by survey responses rather than direct contractual records.
- The exclusion restriction cannot be tested directly.
- Welfare interpretation depends on the measured farmer outcomes and does not capture all possible long-run adjustments.

Interpretation: The paper is strongest as evidence that competition can weaken relational governance in a real supply chain, not as a universal argument against competition.

7 Tables and figures

7.1 Table I and Figure I: summary statistics and relational practices

Read:

- Table I summarizes mill and farmer characteristics.
- The average mill faces about 6.5 competing mills within 10 km and has substantial variation in relational practices.
- Figure I shows that preharvest, harvest, and postharvest relational practices are positively related to the overall relational-contract index.
- The figure validates that the index captures multiple dimensions of a broader relational governance system.

Economic message: Relational contracts are empirically measurable and economically meaningful in the Rwanda coffee chain.

	Mean	25th pct.	Median	75th pct.	Obs
	(1)	(2)	(3)	(4)	(5)
Panel A: Mill characteristics					
Mill age, years	4.090	2	4	6	178
Theoretical capacity (tons of cherries)	423.1	250	340.9	500	173
Production (tons of parchment coffee)	46.01	15	32	60	177
Cherries purchased (tons)	294.8	102.4	199.9	400	174
Seasonal employees	35.13	16	30	50	171
Cooperative status, dummy	0.466	0	0	1	178
Farmers in catchment area that sell to mill	396.0	170	310	500	170
NGO-supported mill, dummy	0.264	0	0	1	178
Total unit cost (RWF per kg)	1,793	1,600	1,800	1,956	178
Total processing unit cost (RWF per kg)	705.3	500.0	699.0	831.0	177
Number of mills within 10 km	6.539	3	6	10	178
Score within 5 km of mill	—	−0.826	−0.276	0.714	177
Score within 5–10 km of mill	—	−0.762	−0.230	0.648	177
Average elevation (m) within 5 km	1,625.8	1,511	1,630.4	1,730.1	177
Average slope (°) within 5 km	10.93	8.859	10.87	12.87	177
Average river density (m) within 5 km	320.5	205.5	319.5	423.2	177
Average tree density within 5 km ('000)	11.53	5.152	9.499	14.64	177
Average spring presence within 5 km	0.033	0.012	0.025	0.049	177
Kilometers of road within 5 km	1.769	1.452	1.674	2.008	177
Overall quality score	—	−0.473	0.150	0.745	159
Given inputs to farmers	0.222	0	0	0	176
Has made a second payment in the past	0.784	1	1	1	176
Provides help/loans to farmers	0.773	1	1	1	176
RC index, mill outcomes	—	−0.894	0.252	0.252	177
RC index, overall	—	−0.502	0.114	0.453	175
Panel B: Farmer characteristics					
Farmer age, years	46.44	36	47	56	875
Female, dummy	0.287	0	0	1	881
Schooling, years	5.339	4	6	7	879
Distance to mill, km	5.480	1.194	2.689	7.182	615
Cooperative membership, dummy	0.552	0	1	1	881
Farmer's trees	975.5	250	500	1,000	881
Cherry price (RWF per kg)	208.2	200	200	220	881
Share sold as cherries (%)	0.792	0.764	1	1	872
Home process for saving, dummy	0.232	0	0	0	881
Job satisfaction index	—	−0.457	0.026	0.499	868
Number of other mills in own quadrant	1.506	0	1	2	615
Received input from mill	0.176	0	0	0	881
Expects to receive a second payment	0.795	1	1	1	881
Expects to receive help/loan	0.637	0	1	1	877
RC index, farmer outcomes	—	−0.659	0.413	0.413	881

Notes. Mill characteristics are obtained from the survey of mills and the authors' GIS data set. Farmer characteristics are obtained from a survey of four or five random farmers supplying to the surveyed mill. Both surveys took place at the same time and were fielded in the harvest season of 2012. Relational-contract index measures, referred to as relational practices in the text, are dummy variables: *Given inputs to farmers*, *Has made a second payment in the past*, and *Provides help/loans to farmers* are responses from mill managers; *Received input from mill*, *Expects to receive a second payment*, and *Expects to receive help/loan* are responses from farmer surveys. Competition is defined as the number of mills within 10 km. Means of standardized

Table I: Summary statistics

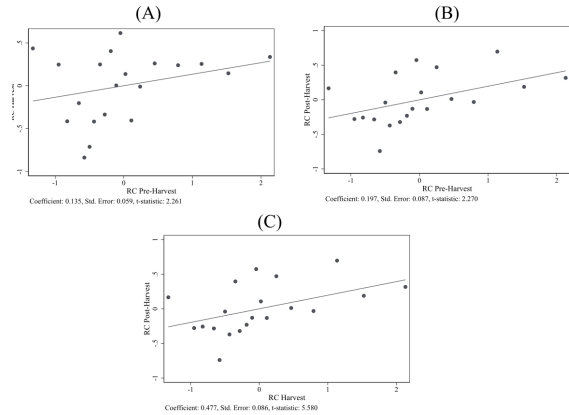


FIGURE I

Relational-Contract Practices

Binned scatter plot of mill-level regressions. All regressions control for mill characteristics (NGO-support, cooperative status, mill age, mill age squared, and

Figure I: relational-contract practices

7.2 Table II and Figure II: competition and relational contracts

Read:

- Table II reports OLS, first-stage, reduced-form, and IV estimates.
- OLS shows that one more competing mill is associated with a 0.116 standard deviation lower relational-contract index.
- The first stage shows that suitability in the 5–10 km donut strongly predicts competition.

- The reduced form is negative: donut suitability predicts lower relational contracting.
- The IV estimate is larger: one additional competitor reduces the relational-contract index by about 0.283 standard deviations.
- Figure II visually reports the first-stage and reduced-form relationships.

Economic message: The IV design suggests a causal negative effect of competition on relational contracting, not just a correlation.

TABLE II
COMPETITION AND RELATIONAL CONTRACTS

Dependent variable	RC index (z-score) (1)	Competition (2)	RC index (z-score) (3)	RC index (z-score) (4)
Competition	-0.116 (0.025)*** <0.025>*** [0.026]***			-0.283 (0.095)*** <0.075>*** [0.060]***
Score within 5–10 km of mill		1.610 (0.329)*** <0.313>*** [0.419]***	-0.455 (0.105)*** <0.107>*** [0.063]***	
Score within 5 km of mill	Yes	Yes	Yes	Yes
Geographic controls	Yes	Yes	Yes	Yes
Mill controls	Yes	Yes	Yes	Yes
Adjusted R^2	0.27	0.70	0.25	0.10
Observations	175	177	175	175
Model	OLS	First stage	Reduced	IV

Notes. Standard errors are denoted as follows: (Bootstrap in which mills are resampled with replacement and the regression is repeated to generate the distribution of the coefficient); <Standard errors adjusted for arbitrary spatial clustering using the *acreg* package written by Konig and coauthors and used in Konig et al. (2017)>; [Standard errors that adjust for spatial clustering as in Conley (1999), implemented by Conley's *xgmm* Stata package]. Stars *** (**) [*] indicate significance at the 0.01 (0.05) [0.1] level. The RC index is a z-score of the three aggregate indices: preharvest, harvest, and postharvest indices with equal weighting of farmer and mill responses. Preharvest z-score is constructed based on farmer- and mill manager-based indicators of mill-provided inputs. Harvest z-score is constructed from farmer- and mill manager-based indicators of promised second payments postharvest. Postharvest z-score is constructed from farmer- and mill manager-based indicators of loans or help provided after the harvest. Competition is measured as the number of mills within a 10 km radius and is instrumented with the engineering model suitability score in locations 5 km to 10 km away from the mill (referred to in the table as "Score within 5–10 km of mill"). For ease of comparison between the OLS and IV estimates, column (1) already includes the average suitability score within 5 km as a control (referred to in the table as "Score within 5 km of mill"). The average suitability scores from the engineering model are all z-scores. Mill controls include whether the mill is NGO-supported, cooperative status, mill age, mill age squared, and mill coordinates. Geographic controls include average engineering suitability score, average spring presence, road density, tree density, rivers, flexible control of FAO-GAEZ coffee suitability, elevation, and slope, all within 5 km of the mill. Coffee suitability is from the FAO's Global

Table II: competition and relational contracts

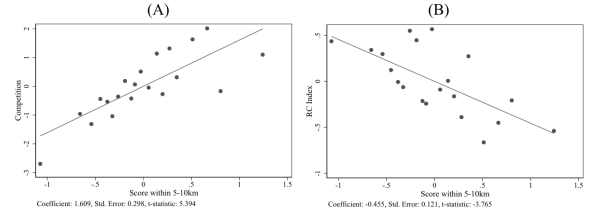


FIGURE II

Instrumental Variable: First Stage and Reduced Form

Binned scatter plot of mill-level regressions. All regressions control for mill characteristics (NGO support, cooperative status, mill age, mill age squared, and mill coordinates). Controls also include average engineering suitability score, average spring presence, road density, tree density, rivers, flexible control of FAO-GAEZ coffee suitability, elevation, and slope, all within 5 km of the mill. The RC index is an aggregate of farmer- and mill manager-based indicators of mill-provided

Figure II: first stage and reduced form

7.3 Tables III and IV: unpacking practices and mill operations

Read:

- Table III decomposes the relational-contract index into preharvest inputs, second payments, credit, loans, and postharvest help.
- Competition reduces many practices at once, supporting the complementarity prediction.
- Table IV studies mill outcomes: total unit costs, cherry prices, home processing, conversion ratio, production, cherry freshness, and labor-related variables.
- Competition raises operational frictions and reduces total cherries processed.

Economic message: Competition breaks the whole relational system rather than simply changing one isolated practice.

Dependent variable	Has made												
	Received input from mill (1)	Given inputs to farmers (2)	RC pre-harvest z-score (3)	Expects a second payment in the past year (4)	RC harvest z-score (5)	Expects to receive help/loans to farmers (6)	Provides help/loans to farmers (7)	RC post-harvest z-score (8)	RC index outcomes (9)	RC index, farmer outcomes (10)	RC index, mill outcomes (11)	Placebo: short-term credit (12)	Placebo: short-term credit (13)
Panel A: IV Competition	-0.064*** (0.014)	-0.085*** (0.036)	-0.220*** (0.112)	-0.063*** (0.017)	-0.077*** (0.035)	-0.203*** (0.074)	-0.066*** (0.020)	-0.026 (0.034)	-0.180*** (0.099)	-0.283*** (0.098)	-0.237*** (0.045)	-0.215*** (0.094)	0.017 (0.082)
Panel B: OLS Competition	-0.011*** (0.005)	-0.039*** (0.012)	-0.062*** (0.022)	-0.038*** (0.006)	-0.041*** (0.013)	-0.121*** (0.038)	-0.021*** (0.007)	-0.015 (0.014)	-0.065*** (0.031)	-0.116*** (0.029)	-0.086*** (0.016)	-0.102*** (0.028)	-0.041 (0.032)
Score within 5 km of mill	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mill controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farmer controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.05	0.13	0.11	0.15	0.16	0.21	0.03	-0.01	0.00	0.10	0.07	0.12	0.04
Observations	869	176	176	869	176	176	865	175	175	175	869	176	172

Notes. Bootstrapped standard errors are in parentheses. *** (**) (+) indicates significance at the 0.01 (0.05) (0.1) level. RC preharvest z-score is constructed based on farmer- and mill manager-based indicators of mill-provided inputs. RC harvest z-score is constructed from farmer- and mill manager-based indicators of promised second payments postharvest. RC postharvest z-score is constructed from farmer- and mill manager-based indicators of loans or help provided after the harvest. The RC index (column 10) is an aggregate of these three indices with responses from the farmer and the mill manager equally weighted. Aggregate relational practices from the farmer's perspective and mill manager's perspective are reported in columns (11) and (12), respectively. Column (13) captures if there was any short-term credit provided to the farmer by the mill. Mill controls include NGO-supported, cooperative status, mill age, mill age squared, and mill coordinates. Geographic controls include average engineering score, average spring presence, road density, tree density, rivers, flexible control of FAO-GAEZ coffee suitability, elevation, and slope, all within 5 km of the mill. Farmer controls include farmer age, education, gender, schooling, distance to mill, cognitive test (z-score), and cooperative membership. Farmer responses are from the 2012 farmer survey. Competition is measured as the number of mills within a 10 km radius and is instrumented with the engineering model suitability score in locations 5 km to 10 km away from the mill. Adjusted R² is provided for Panel A (IV). For additional variable definitions refer to notes in Table II.

Table III: unpacking relational practices

7.4 Tables V and VI: quality and farmer outcomes

Read:

- Table V shows that competition reduces overall coffee quality and farmer-controlled quality but does not significantly affect purely mill-controlled quality or plant genetic properties.
- Table VI shows effects on farmer outcomes including prices, home processing, sales, self-financed inputs, revenues, wages, job satisfaction, and other welfare proxies.
- The results support the view that competition weakens relational services and makes farmers worse off.

Economic message: The breakdown of relational contracting has real consequences for quality, mill performance, and farmer welfare.

Dependent variable	Overall quality score (1)	Farmer-controlled quality (2)	Mill-controlled quality (3)	Plant genetic properties (4)
Panel A: IV Competition	-0.146* (0.087)	-0.172* (0.089)	-0.034 (0.085)	0.018 (0.062)
Panel B: OLS Competition	-0.039 (0.034)	-0.051* (0.031)	-0.043 (0.037)	0.024 (0.021)
Score within 5 km of mill	Yes	Yes	Yes	Yes
Engineering controls	Yes	Yes	Yes	Yes
Geographic controls	Yes	Yes	Yes	Yes
Mill controls	Yes	Yes	Yes	Yes
Adjusted R ²	-0.075	0.040	0.19	-0.013
Observations	158	155	156	157

Notes. Bootstrapped standard errors are in parentheses. *** (**) (+) indicates significance at the 0.01 (0.05) (0.1) level. Column (1) is the overall quality score constructed from the farmer and mill quality indices with equal weighting to all indicators, plus an indicator of the ideal conversion ratio, an indicator of specialty status, and standardized cupping points. Column (2) is the farmer-controlled quality outcome as a standardized index of an indicator of large beans and severe insect damage. Column (3) is the mill-controlled quality index constructed from an indicator of high moisture, floaters, and broken beans. All components of indices are rescaled so that higher values indicate higher quality. Mill controls include NGO-supported, cooperative status, mill age, mill age squared, and mill coordinates. Engineering controls and geographic controls include average engineering suitability score, average spring presence, road density, tree density, rivers, flexible control of FAO-GAEZ coffee suitability, elevation, and slope, all within 5 km of the mill. Competition measure is the number of mills within a 10 km radius, and is instrumented with the engineering model suitability score in locations 5 km to 10 km away from the mill. Adjusted R² is provided for Panel A (IV). For additional variable definitions refer to notes in Table II.

Table V: competition and quality of output

Dependent variable	Total unit cost (RWF, ln) (1)	Cherry price (RWF, ln) (2)	Home-processed purchase price (RWF, ln) (3)	Conversion ratio (ln) (4)	Total processing unit cost (RWF, ln) (5)	Cherries purchased (ton) (6)	Weeks mill processed (7)	Days with too many workers (8)	Days with too few workers (9)	Unit labor costs (10)
Panel A: IV Competition	0.046* (0.026)	0.013 (0.013)	-0.007 (0.011)	-0.007 (0.016)	0.071*** (0.031)	-4.884* (2.877)	-0.202 (0.322)	0.227** (0.099)	0.095* (0.055)	0.107* (0.058)
Panel B: OLS Competition	0.010* (0.005)	0.002 (0.004)	-0.001 (0.003)	-0.011*** (0.004)	0.017 (0.011)	-4.733 (0.931)	0.009 (0.142)	0.107*** (0.032)	0.044* (0.026)	0.062*** (0.019)
Score within 5 km of mill	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Engineering controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mill controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	-	0.24	0.28	-0.02	-0.08	0.24	0.08	0.13	0.01	0.16
Observations	177	163	176	145	176	169	160	173	174	176
Dep. var. mean	7.48	5.34	6.64	4.17	6.51	27.17	15.11	2.20	1.29	5.44

Notes. Bootstrapped standard errors are in parentheses. *** (**) (+) indicates significance at the 0.01 (0.05) (0.1) level. Column (1) is the total unit cost of producing 1 kg of the output (purchase). Mill manager's response to price paid to farmers for cherries (ln) and local home-processed purchase price (ln) are reported in columns (2) and (3) respectively. Column (4) is the physical efficiency ratio between the input (cherry) and output (purchase). Column (5) reports the processing cost (excluding cost of cherry) from the unit cost of processing reported in column (1). Column (6) is the total cherries purchased by the mill as reported to the coffee board in the 2012 season. Mill capacity is controlled for in this specification. Column (7) is the number of weeks the mill is operational in the 2012 season. In column (8) and (9) the dependent variables, "Days with too many workers" and "Days with too few workers" are all dummy variables reported by the mill manager. Competition is measured as the number of mills within 10 km and is instrumented with the engineering model suitability score in locations 5 km to 10 km away from the mill. Mill controls include NGO-supported, cooperative status, mill age, mill age squared, and mill coordinates. Engineering controls and geographic controls include average engineering score, average spring presence, road density, tree density, rivers, flexible control of FAO-GAEZ coffee suitability, elevation, and slope, all within 5 km of the mill. Adjusted R² is provided for Panel A (IV). Responses are provided by the mill manager. For additional variable definitions refer to notes in Table II.

Table IV: competition and mill outcomes

Dependent variable	Cherry price (RWF, ln) (1)	Home-processed purchase price (RWF, ln) (2)	Share sold as cherries (3)	Home process for saving (4)	Self-financed inputs (5)	Yield (ln) (6)	Input usage (RWF, ln) (7)	Farmer's revenues (RWF, ln) (8)	Farmer Job satisfaction index (9)	Freedom to decide how to do the work (10)	Pay is good (11)	Work is stressful (12)	Work is successful (13)
Panel A: IV Competition	0.012*** (0.003)	-0.004 (0.004)	-0.075*** (0.017)	0.059*** (0.020)	0.034* (0.019)	-0.004 (0.032)	-0.024 (0.063)	0.057 (0.036)	-0.081* (0.048)	-0.059** (0.024)	-0.065* (0.029)	-0.047* (0.029)	0.078*** (0.023)
Panel B: OLS Competition	0.004*** (0.001)	0.007*** (0.001)	0.003 (0.006)	0.007 (0.006)	0.025*** (0.006)	0.039*** (0.010)	-0.016 (0.021)	0.017 (0.015)	0.083*** (0.014)	-0.035*** (0.009)	-0.030*** (0.011)	-0.035*** (0.009)	-0.005 (0.010)
Score within 5 km of mill	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Engineering controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mill controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Farmer controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.26	0.32	-	0.00	0.08	0.14	0.06	0.14	0.36	0.12	0.11	0.07	-0.02
Observations	869	869	860	869	865	472	869	848	856	855	855	855	855
Dep. var. mean	5.33	6.63	0.79	0.23	0.35	0.38	1.20	6.22	4.84	-0.00	2.96	1.61	3.48

Notes. Bootstrapped standard errors are in parentheses. *** (**) (+) indicates significance at the 0.01 (0.05) (0.1) level. Column (1) reports farmers' cherry price and column (2) reports local home-processed purchase price. Column (3) reports share of farmers' coffee output sold as cherries to mills. Column (4) is the farmers' response to an open-ended question on why they would home process coffee, with a dummy variable created for the responses that state saving as the main motive. Column (5) is a dummy variable indicating if the farmer self-financed input purchases for their plot. Column (6) is the yield (ln ln), and column (7) is the amount of self-financed inputs per kg of output (cherries). Column (8) is the farmer's tree holdings (ln), and column (9) is the farmer revenues (sum of revenues received from coffee sold as cherries and as home-processed purchase). Column (10) is a z-score of 1-4 (4 being strongly agree) responses of statements: "My job gives me a chance to do the things I do best," "The pay is good," "I learn new things," "I am treated with respect," "I have freedom to do my work," and "I do not find work stressful." Columns (11)-(13) are some of the individual components of the job satisfaction index. Mill controls include NGO-supported, cooperative status, mill age, mill age squared, and mill coordinates. Engineering controls and geographic controls include average engineering score, average spring presence, road density, tree density, rivers, flexible control of FAO-GAEZ coffee suitability, elevation, and slope, all within 5 km of the mill. Competition is measured as the number of mills within 10 km, and is instrumented with the engineering model score in locations 5-10 km away from the mill. Farmer controls include farmer age, education, gender, schooling, distance to mill, cognitive score, cooperative membership, and log number of coffee trees (except in column 7). All farmer outcomes/responses are from the 2012 farmer survey. Adjusted R² is provided for Panel A (IV).

Table VI: competition and farmer outcomes

7.5 Table VII: direct and indirect mechanisms

Read:

- Table VII separates baseline competition, reduced-sample specifications, distance to nearest competitor, and farmer/mill-level access measures.

- Farmer-level competition and all-level competition help explain farmer outcomes and relational-contract outcomes.
- The decomposition supports both the direct temptation channel and the indirect mill-profit channel.

Economic message: Competition matters not only because farmers can sell elsewhere, but also because it reduces the focal mill’s ability to sustain relationship rents.

TABLE VII
RELATIONAL CONTRACTING AND COMPETITION: FARMER LEVEL

	Baseline (1)	Reduced sample (2)	Distance to nearest mill (3)	Distance to second-nearest mill (4)	BC index, farmer outcomes (5)	Farmer competition (6)	Mill competition (7)	BC index, farmer outcomes (8)
Competition	-0.257*** (0.047)	-0.213*** (0.053)	-0.265*** (0.057)	-0.269*** (0.055)				
Farmer mill access			-0.042* (0.022)	-0.083** (0.042)				
Farmer-level competition					-0.055** (0.025)			-0.342*** (0.125)
Mill-level competition					-0.103*** (0.017)			-0.223*** (0.085)
Engineering score in own quadrant					0.870*** (0.104)			-0.579*** (0.136)
Engineering score in other quadrants					-0.274*** (0.106)			1.417*** (0.228)
Equality of Coefficients, <i>p</i> -value								101
Competition definition								# mills in quadrant (farmer), # mills in other quadrants (mill)
Instrument								Score in quadrant, score in other quadrants

Table VII: relational contracting, farmer level

TABLE VII
CONTINUED

	Baseline (1)	Reduced sample (2)	Distance to nearest mill (3)	Distance to second-nearest mill (4)	BC index, farmer outcomes (5)	Farmer competition (6)	Mill competition (7)	BC index, farmer outcomes (8)
Mill controls			Yes	Yes	Yes	Yes	Yes	Yes
District fixed effects					Yes	Yes	Yes	Yes
Farmer controls					Yes	Yes	Yes	Yes
Quadrant fixed effects					Yes	Yes	Yes	Yes
OLSV					0.85	0.85	0.85	IV
Adjusted R ²	0.07	0.11	0.13	0.12	0.18	0.47	0.64	-0.00
Observations	869	696	696	696	696	696	696	696

Notes: Bootstrapped standard errors are in parentheses. *** (**) (*) indicates significance at the 0.01 (0.05) (0.1) level. The dependent variable across all columns is the relational-contract index for farmer outcomes. Column (1) includes “farmer mill access”, this is a measure of a farmer’s access to a mill, analogous to Donohue and Hershock (2016); the measure is constructed as the inverse sum of the distances to the nearest and second-nearest competing mills to the farmer with no restriction on the radius. In column (4) the “farmer mill access” measure is now the inverse of the distance to the second-nearest competing mill with no restriction on the radius. Farmer controls include distance to mill, farmer age, education, gender, schooling, cognitive score, cooperative membership, and holdings of coffee trees (a). Mill controls include NGO-supported, cooperative status, mill age, mill age squared, and mill coordinates. Additional geographic controls include the average engineering suitability score, average spring pressures, road density, tree density, rivers, Euclidean control of FAO GAEZ coffee suitability, elevation, and slope, all within 5 km of the mill. Quadrants are adjacent 10 km by 10 km squares with the mill situated at the shared corner. Quadrant fixed effects refer to dummies for the northeast, southeast, and so on, quadrant. Many farmer outcomes included in the relational-contract index measure are receiving fertilizer from the mill, receiving a second payment, and expecting a loan or help after harvest. The number of observations falls when quadrant-level competition is introduced because we only have exact village location information for 70% of our surveyed farmers in 2012. *p*-value is reported for the test that farmer and mill effects are equal.

Table VII continued

8 Detailed result synthesis

8.1 Competition reduces relational contracting

The most important result is that competition lowers the relational-contract index. The IV estimate is more than twice the OLS estimate, consistent with measurement error in the competition variable or omitted-variable bias that attenuates the OLS relationship. The magnitude is economically meaningful: one more competing mill reduces relational contracting by about 0.28 standard deviations.

8.2 The practices are complementary

The decline is not confined to one dimension. Competition reduces input provision, credit, delayed payment, and postharvest assistance. This is exactly what a relational-contract model predicts if multiple practices are sustained by the same future relationship surplus.

8.3 Mills perform worse

Competition increases operational frictions. When relational sourcing breaks down, mills receive more irregular deliveries, face planning problems, process fewer cherries, and produce lower quality coffee. Competition can therefore reduce productive efficiency even though it increases buyers’ contest for farmers.

8.4 Farmers can be worse off

The farmer result is the strongest challenge to a simple competition-is-good story. Although competition might improve spot outside options, farmers lose inputs, credit, delayed payments, and assistance. In this environment, those relationship benefits dominate, so farmers become worse off.

8.5 Direct and indirect channels both matter

The farmer-level evidence supports two channels. Competition directly tempts farmers to default on the relationship, especially when alternative mills are nearby. Competition also reduces the focal mill’s profits, which weakens its willingness and ability to provide relational services.

9 Short summary

- The paper studies competition and relational contracts in Rwanda’s coffee chain.

- Mills and farmers rely on informal agreements because formal enforcement is weak.
- Relational practices include inputs, loans, delayed payments, and postharvest assistance.
- Competition is measured as the number of mills within 10 km.
- The instrument uses engineering suitability for mill placement in the 5–10 km donut around the focal mill.
- One more competing mill reduces the relational-contract index by about 0.28 standard deviations in IV estimates.
- Competition weakens all major relational practices simultaneously.
- Competition worsens mill operations and reduces coffee quality.
- Competition reduces the aggregate amount of coffee supplied to mills and makes farmers worse off.
- The mechanism combines a direct farmer temptation channel and an indirect mill-profit channel.
- The paper shows that in weak-contracting environments, competition can be excessive and socially harmful.

10 Conclusion

10.1 Core conclusion

Macchiavello and Morjaria show that competition can undermine relational contracts in markets where formal enforcement is weak. In Rwanda’s coffee sector, additional competing mills reduce relational practices, lower mill performance, reduce coffee quality, and make farmers worse off.

10.2 Economic intuition

Relational contracts require future surplus to discipline current behavior. Competition raises outside options and lowers mill profits. Both effects reduce the future value of relationships, making it harder for mills and farmers to sustain input provision, credit, delayed payments, and postharvest assistance.

10.3 Takeaway

The paper’s broader lesson is that market competition is not always efficiency enhancing in second-best environments. When informal relational governance is essential, policies that increase entry or competition may unintentionally destroy the relational institutions that make exchange work.